Gyutae Oh

CURRICULUM VITAE

Media System Lab (MSL), Sungkyunkwan University
Electrical and Computer Engineering
alswo740012@g.skku.edu
Portfolio
Google Scholar
Git
LinkedIn

AI graduate researcher focused on **efficient learning**, **continual learning**, **anomaly detection**, and **medical image analysis**. Currently pursuing a Ph.D. in Electrical and Computer Engineering at Sungkyunkwan University under the supervision of Prof. Jitae Shin.

EDUCATION

Sungkyunkwan University, Ph.D. in Electrical and Computer Engineering — Feb 2026 – Present

Sungkyunkwan University, M.S. in Electrical and Computer Engineering — Feb 2024 – Feb 2026

GPA: 4.07/4.5

Korea National University of Transportation, B.S. in Computer Engineering — Mar 2019 – Feb 2024

GPA: 3.79/4.5

RESEARCH EXPERIENCE

Media System Lab (MSL), Sungkyunkwan University — Jan 2024 – Present

Advisor: Jitae Shin

Applied Machine Intelligence Lab (AMI-Lab), KNUT — Mar 2022 – Jun 2023

Advisor: Jeonghwan Gwak

MILITARY SERVICE

RESEARCH INTERESTS

- Continual Learning
- Efficient Learning
- Anomaly Detection
- Medical Image Analysis
- PyTorch
- Evaluation

RESEARCH OUTPUTS

11 total papers

4 preprints · 7 peer-reviewed publications

1 patent

7 scholarships · 4 awards · 1 certification · 1 service

HONORS & AWARDS

Hanium: Companion Dog Eye Disease Detection Assistant App — Selected Works Prize (2023.11)

Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Republic of Korea Army, Sergeant, Honorably Discharged

2020 – 2022

TOD operator at Gachilbong GOP, 21st Infantry Division, under a direct unit of III Corps

SELECTED PUBLICATIONS

† First author, * Corresponding author

Routing Prompts Across Layers: A Global Memory Module for Prompt-Based Continual Learning

Gyutae Oh, Jitae Shin

International Conference on Computer Research and Development (ICCRD), January 24, 2026

Multimodal Fusion Framework Using Contrastive Learning for Exposure Keratopathy

G Oh†, Y Won†, D Lim, J Shin*

International Workshop on Ophthalmic Medical Image Analysis (MICCAI 2025 Workshop), 2025

Paper Code

Patch Core++: Augmenting Coreset for Superior Industrial Anomaly Detection

G Oh, J Shin*

IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia), 2024

Paper

Feature Ensemble-based Wolff Parkinson White Syndrome classification through ECG

G Oh†, I Kim, B Kim, Y Jeon, J Gwak*

Proceedings of the Korean Society of Computer Information Conference, 2023

Paper

UNDER REVIEW

AAAI 2027

1 anonymous submission under review

JBHI 2026

1 anonymous submission under review

WACV 2027

1 anonymous submission under review

ACCV 2026

Companion Dog Eye Disease Detection Assistant App — Silver Prize & Graduation Project (2023.08)

Korea National University of Transportation

Video

Hanium: Automated Serving Robot using Object Detection & SLAM — Bronze Prize (2020.08)

Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Video

Campus Contest: Automated Serving Robot using Object Detection & SLAM — Grand Prize (President's Award) (2020.09)

Korea National University of Transportation

Video

SCHOLARSHIPS

SKKU BK21 Fellowship (Specialized Scholarship) (2024)

Sungkyunkwan University

University Innovation — Merit & Living Expense Scholarship (2024)

Sungkyunkwan University

Academic Excellence Scholarship (awarded 5 times, 2019–2023)

1 anonymous submission under review

SELECTED PROJECTS

Exposure Keratitis Diagnosis & Severity Prediction (Year 2)

2025.10 – Present

Sungkyunkwan University · Kangbuk Samsung Hospital collaboration

Extending multimodal medical AI research toward model compression and clinically practical deployment.

AI CHAMPION

2025.07 – 2025.09

Fundus-image-based multi-disease screening project with cross-lab and hospital collaboration; selected as a Top 100 team out of approximately 500 teams.

Exposure Keratitis Diagnosis & Severity Prediction (Year 1)

2024.10 – 2025.10

Developed multimodal AI models for exposure keratitis diagnosis and severity prediction from anterior segment photographs, leading to a workshop publication and a domestic patent application.

Korea National University of Transportation

PATENT

Device and Method for Diagnosing Exposure Keratopathy based on Multimodal Fusion Learning

Application No. 10-2025-0121342 · Filed Aug 28, 2025

Two-stage multimodal fusion learning framework for automatic grading of Exposure Keratopathy (EK). Trained with four ocular imaging modalities collected at a tertiary hospital (broadbeam, slit-beam, scattering, and blue-light): modality-wise features are aligned and fused during training, while inference enables modality-agnostic classification using only a single broadbeam image to implicitly leverage information from other modalities.

CERTIFICATION

Advanced Data Analytics Semi-Professional (ADsP)

Issued by Korea Data Agency (KDATA) · Sep 2025

ID: ADsP-046012145

PROFESSIONAL SERVICE

Reviewer (2026)

IEEE International Conference on Image Processing (ICIP)

Updated from portfolio content on March 19, 2026.